

Problem Statement

The Challenge

Places change detection for map data maintenance

Sources – OSM

- OSM provides community-curated Point-of-Interest (POI) data representing businesses and public locations such as restaurants, retail stores, hospitals, fuel stations, etc.
- It serves as the baseline reference dataset for existing places in the map database.
- Unlike real-world signals (e.g., delivery platforms, social media, business listings), OSM may lag behind actual business openings or closures.
- Places are typically represented as POINT geometries (sometimes polygons for larger venues).
- Coordinates are provided in WGS84 (EPSG:4326).

	osm_id	type	latitude	longitude	name	category \
0	368261001	node	1.389365	103.886287	None	cafe
1	380582666	node	1.359490	103.885485	Suki-ya	restaurant
2	393149179	node	1.331466	103.948238	None	cafe
3	462263947	node	1.305148	103.930677	Jumbo Seafood	restaurant
4	462264013	node	1.289259	103.848273	Jumbo Seafood	restaurant
	cuisine		street		postcode	
0	coffee_shop		None		None	
1	japanese		Hougang Street 21		530205	
2	coffee_shop		None		None	

Attributes

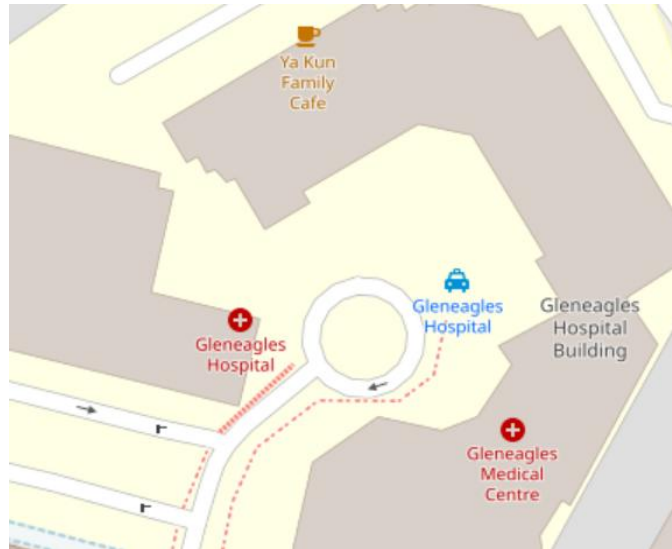
- Place ID (OSM ID)
- Place Name
- Category Tag (e.g., amenity, shop, tourism)
- Place Type (e.g., restaurant, cafe, supermarket, hospital)
- Address Information (street, city, postcode)
- Contact Details (phone, website)
- Operational Info (opening hours)
- Metadata (last edit timestamp, source tag)

Geometry Format
Delivered primarily as POINT geometries
Coordinates in WGS84 (EPSG:4326)



The Challenge

OSM POI examples



The challenge focuses on detecting place changes only within the following categories: **restaurants, cafes, tourist attractions, hotels, amusement parks, shopping malls, department stores, grocery stores, pharmacies and fuel stations**, mapped to corresponding **OSM tags (amenity, shop, tourism, leisure)** **without the use of Google (API/ Search Engine or any related source)**

Real-time Examples

Place Name: Cafe Mary Grace
Address: 52 Tras Street, #01-01
Category: Restaurant
Coordinates: 1.2781321,103.8440058
Match in Database: Not Found wrt OSM
Predicted Status: New Place
Confidence: 0.87
Confirmed From: food delivery apps, social media check-ins
Recommendation: Accept (Review / Reject / Accept)

Place Name: Punggol Coast Mall
Address: 88 Punggol Way, Singapore 829913
Category: Shopping Mall
Coordinates: 1.415708,103.9105519
Match in Database: Not Found wrt OSM
Predicted Status: New Place
Confidence: 0.6
Confirmed From: blogs, social media check-ins
Recommendation: Review (Review / Reject / Accept)

Place Name: The Laurus, a Luxury Collection Resort, Singapore
Address: 28 Sentosa Gateway Singapore 098139
Category: Hotel
Coordinates: 1.2579631,103.8162503
Match in Database: Not Found
Predicted Status: New Place
Confidence: 0.9
Confirmed From: blogs, social media check-ins
Recommendation: Accept (Review / Reject / Accept)

Place Name: Cnergy Station at Dunman Road
Address: 743 Dunman Rd, Singapore 439241
Category: Fueling Station
Coordinates: 1.3097546430974762, 103.90092030072083
Match in Database: Not Found
Predicted Status: New Place
Confidence: 0.8
Confirmed From: blogs, social media check-ins
Recommendation: Accept (Review / Reject / Accept)

Evaluation Criteria

Category	Points	Description
Innovation & Creativity	10	Evaluates how unique and creative the proposed solution is. Includes originality in using alternative data sources (e.g., delivery platforms, social signals, business directories, street imagery) and innovative methodologies for detecting new or closed places without relying on restricted APIs.
Technical Realization	10	Assesses how effectively the solution uses relevant technologies such as AI/ML models, geospatial processing, data pipelines, APIs, and databases. Judges' technical depth, correctness of implementation, reproducibility, and whether the prototype works reliably.
Scalability & Business Value	5	Measures how practical and scalable the solution is for real-world deployment. Considers automation capability, ability to handle large geographic datasets, and potential value for map maintenance, navigation systems, and location intelligence platforms.
Presentation & Communication	5	Evaluates clarity of explanation, structure of the presentation, ability to communicate technical concepts effectively, and how well the team demonstrates the impact and feasibility of their solution.

Deliverables

- Working prototype (code or pipeline demo)
- Architecture diagram
- Model evaluation results

Example 1:

Place Name: Marina Bay Noodle House

Address: Marina Bay (dummy address)

Coordinates: 1.2834, 103.8607

Match in Database: Not Found

Predicted Status: New Place

Confidence: 0.87

Confirmed From: food delivery apps, social media check-ins

Recommendation: Review / Reject / Accept

Place Name: Orchard Street Café

Address: Orchard Road (dummy address)

Coordinates: 1.3048, 103.8318

Match in Database: Found

Predicted Status: Closed

Confidence: 0.90

Confirmed From: Business listing updates, user reviews
mentioning closure, street imagery updates